

Casual Riders on Bike Share Toronto Are Not All Leisure Riders*

In 2025, casual riders' weekday trips look similar to those of members during the winter, while their summer trips look like leisure

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I use all 7.8 million Bike Share Toronto trips from 2025 to compare members and casual riders on when trips start, duration of their trips, and how these patterns change throughout the year. Casual trips look like leisure across the year, as 11% of casual weekday trips start during the morning rush, compared to 19% for members. In the winter, 17% of casual weekday trips start in the morning rush, nearly double the 9% from the summer. 'Casual' appears to include two groups: a small year-round group that rides like members on weekdays, and a large summer group that rides for leisure, so treating all casual riders as leisure riders misses a part of how the bike share system is used.

1 Introduction

Bike Share Toronto is a fast-growing bike share system. It grew from 780+ stations and a projected 5.5 million rides in 2023 (Toronto Parking Authority 2023), to over 7.8 million rides in 2025 (Toronto Parking Authority 2026b). It is one of the few North American bike share systems within a cold climate that runs all throughout the year (El-Assi et al. 2017). Within Bike Share Toronto, riders are recorded as either members, or casual riders.

A common picture we see is that bike share members commute, whereas casual riders ride for leisure. Buck et al. (2013) ran a survey, where most short-term users were visitors riding for tourism purposes in Washington DC. El-Assi et al. (2017) supported the idea that members commute, but casual riders ride for leisure or utilitarian activities within Toronto. However, El-Assi et al. (2017) used old 2013 data, when the bike share only had 80 stations. By 2025, the system had grown almost tenfold. Pricing also changed in 2023, allowing casual riders to

*Code and data are available at: <https://github.com/harryyz/toronto-bike-share-2025>.

pay per ride. The publisher of the Bike Share Toronto data defined “casual” only for their 2014-2016 data, before these massive changes and system growth (Toronto Parking Authority 2026b). Thus it is unclear whether “casual” riding still means leisure, or if there has been a change in meaning.

With all 7.8 million trips from 2025, I compare members and casual riders on the hour they start their trips, the duration of their trips, and how these patterns change throughout the year. Firstly, I find that across the whole year, casual trips do look like leisure. Looking at the timing of when rides start, 11% of casual weekday trips start during the morning rush between 7 and 10 am, while 19% of members start their trips during this same rush. The median casual trip is 17.8 minutes on the weekends, but only 10.7 for members. Whereas during weekdays, this gap is much smaller, 13.8 minutes for casual riders and 10.2 for members. Secondly, casual riding is far more seasonal than member riding. The number of casual trips grew by 45 times by its peak from February to August, while member trips grew only 7 times by its peak from February to July. Thirdly, “casual” actually looks like two subgroups of riders. During winter, January and February, 17% of casual weekday trips began during the morning rush, which was almost double the 9% in the summer, July and August. This statistic changed a lot less for members, with 21% in the winter and 18% in the summer. There appears to exist a small group of year-round casual riders who ride much like members during the morning commute. The other group of casual riders is much larger, and appears to ride for leisure during the summer.

If we treat all casual riders as just riders for leisure, we are missing an entire subgroup of casual riders who ride throughout the whole year, and demonstrate similar weekday riding patterns to members. This may matter for the city and the bike share operator when they analyse ridership figures by ridership group to perform tasks like plan winter service. This “member-like” group existing may also be important to consider when pricing memberships. One caveat is that this data records trips and not riders, so all patterns I discuss regard trips and not individuals.

The remainder of the paper is arranged as follows. Section 2.1 discusses the data source, context, and considerations. Section 2.2 describes how a ride becomes a record, and what “casual” means in the data. Section 2.3, Section 2.4, Section 2.5, and Section 2.6 are numerous graphical representations of the data. Section 2.7 contains a summary statistics table. Section A and Section B detail data cleaning and tests.

2 Data

2.1 Source, Context, and Considerations

Bike Share Toronto Ridership Data is published by the Toronto Parking Authority (TPA) on the City of Toronto Open Data Portal (Toronto Parking Authority 2026b). I access this dataset through `opendatatoronto` (Gelfand 2025). The data is refreshed monthly, but published as one zip file per calendar year, with each yearly zip containing 12 monthly csv files.

This paper uses all 12 files from 2025, containing a total of 7,812,520 trips. After data cleaning, I was left with 7,788,842 trips. Each row of data represents one trip, with the following fields: trip ID, duration, start and end station, each with an ID and name, start and end time, bike ID, type of user, which is either member or casual, and bike model (Table 1).

This data is collected automatically by the bike share system as part of its operation and customer charging process. TPA, which runs this system, is a self-funding city agency, and uses parking revenue to fund expansion of bike share. The system has grown from 80 stations and ~400,000 rides in 2011, to 780+ stations and a projected 5.5 million rides in 2023 (Toronto Parking Authority 2023). This data of 7.8 million trips in 2025 suggests that ridership has grown even further since 2023.

I selected 2025 as a useful year because a new rate structure was launched on April 3, 2023, which was the first change since 2017 (Toronto Parking Authority 2023). The casual prices during the new rate structure launch in 2023 match those listed on the current pricing page as of September 2026 (Cycle Toronto 2023; Bike Share Toronto 2026), suggesting that casual pricing was broadly stable throughout my study period. Furthermore, the Ontario government passed a law in 2024 to remove bike lanes on Bloor Street, Yonge Street, and University Avenue, and a lower court blocked removal in July 2025. The Court of Appeal reversed this decision on August 14, 2026 (Harrison and Silva Ponte 2026). Thus, data from 2025 is a snapshot before any removal occurred. This paper does not measure the effect of major bike lane changes because they did not occur in 2025.

On ethical considerations, all trip data is anonymized, where no rider names or account IDs are provided so that riders cannot be linked across separate trips. This protects user privacy, but also means that I am only able to study patterns on trips, not people. However, it should be mentioned that patterns in exact start and end times and stations could theoretically help identify likely users who regularly travel the same route. Regarding coverage, stations are concentrated downtown, but the TPA is working to expand into Neighborhood Improvement Areas. For instance, 211 of 375 new stations in 2024 were established within these areas (Toronto Parking Authority 2023). The data describes people with access to stations, which is not necessarily all Toronto residents.

On statistical considerations, our dataset is a census of every trip in 2025, not a sample, so the counts have no sampling error. Because the population is trips and not riders, and frequent riders may contribute many trips, patterns in the data are weighted towards frequent users. For example, a member using the service to commute will ride 400 times and be counted 400 times, whereas a tourist who rides once will be counted once. I also cannot extrapolate our results to all cycling in Toronto, because I must exclude people riding their own bikes in the city.

A similar dataset I considered was the Bike Share Toronto real-time station feed, which provides real-time bike and dock availability. However, this dataset currently keeps no history, and shows station status rather than trips. There is currently a public request to “track historical bikeshare station locations as they change” (Toronto Parking Authority 2026a). I also considered using

Table 1: Variables used in the analysis. Each row of the data is one Bike Share Toronto trip taken in 2025. Constructed variables were derived from the recorded fields during cleaning.

Variable	Description	Source	Example
Start and end time	Local date and time the bike was unlocked and returned	Recorded	2025-07-14 08:15
Trip length	Minutes between unlocking and returning the bike	Recorded	9.6
Start and end station	Docking station where the trip began and ended	Recorded	Bay St / Wellesley St W
Rider type	Member or casual; see the Measurement section	Recorded	Member
Bike type	Classic pedal bike or e-bike	Constructed	Classic
Month	Month the trip began	Constructed	July
Day type	Weekday, or weekend or Ontario statutory holiday	Constructed	Weekday
Start hour	Hour of the day the trip began (0 to 23)	Constructed	8
Round trip	Whether the trip ended at the station where it started	Constructed	No

earlier years of our current dataset (Toronto Parking Authority 2026b), but I determined that 2025 was a good year with the most data points that capture the new rate structure.

I used the statistical programming language R for all analysis (R Core Team 2026). I used `ckanr` to locate the download link in the download script (Chamberlain et al. 2023). I used `tidyverse` for data cleaning (Wickham et al. 2019), and `arrow` to save and read the data (Richardson et al. 2026). I used `testthat` for data tests (Wickham 2026). I used `ggplot2` to generate figures (Wickham 2016), and `tinytable` for tables (Arel-Bundock 2026). I used `here` for file paths (Müller 2025), and `knitr` for Quarto to run the code (Xie 2015).

2.2 Measurement

Each row in the dataset is created automatically as a rider unlocks and returns a bike. First, a rider unlocks a bike at a station using a membership, day pass, or pay-as-you-go selection. The system automatically records the trip ID, start time, start station, and bike ID at the

moment of unlock. When the rider docks the bike to any station after their ride, the system then records the end time and end station. The duration is calculated as the time between unlocking and bike return, in seconds. User type is set according to how the rider paid, as a “Member” or a “Casual”. The bike model is recorded according to the type of bike used in the process, either an ICONIC which is a classic bike, or an EFIT/EFIT G5 which is an e-bike.

What this dataset fails to measure is the route taken with the bike, or the total distance traveled. Whether the purpose of the trip was for commuting or leisure is also not measured, but may be inferred from timing and duration of the ride, and whether it is a round trip. Therefore, mentions about “commuting” in this paper are not measured detail, but interpretations of patterns from the data.

It was difficult to define who the “Casual” label actually covered. The publisher’s readme defines members as “annual pass holders” and casual users as “24 or 72 hour pass holders” (Toronto Parking Authority 2026b). But this readme is written to cover files from 2014-2016. In 2023, bike share pricing changed, and the 72-hour pass was removed, while single rides became pay-as-you-go priced per minute (Toronto Parking Authority 2023; Cycle Toronto 2023). After restructuring, non-membership products now include pay-as-you-go and the Day Pass (Bike Share Toronto 2026). In 2025, “Casual” most likely covers both day-pass holders and pay-as-you-go users because these are the only two non-membership products that the operator offers. When the data records “Casual”, we do not know if the rider used a day pass or pay-as-you go, and also do not know whether they are a resident or visitor. This paper examines whether casual trips look like leisure or commuting, using data such as when they start, their duration, whether they are a round trip, and how all of this changes by season.

Pricing shapes what we measure. Under current pricing, annual memberships include unlimited 30- or 45-minute classic trips depending on tier. Exceeding the time limit costs extra, so members have a reason to limit their trips to their tier’s limit (Bike Share Toronto 2026). Figure 3 shows member trips dropping sharply after 30 and 45 minutes, while casual trips do not. The day pass allows unlimited 90-minute classic rides, which is friendly for long leisure rides. Differences in trip length can reflect not just trip purpose, but pricing differences as well.

To fix recording problems in the data, I removed 23,678 trips (0.3%), including trips that were missing an end station or end time, trips under a minute likely due to failed unlocks, and trips over a day likely due to system error. I also noticed that from January to October, all end-station names erroneously repeated the start station’s name, which I fixed by rebuilding the names from the station IDs, which were listed correctly. Detailed data cleaning is described in Section A.

2.3 Rider type and season

Members made 72.6% of the trips in 2025, while casual riders made 27.4% (Figure 1). Though both groups rode more in the summer, casual riding was more seasonal. From February to

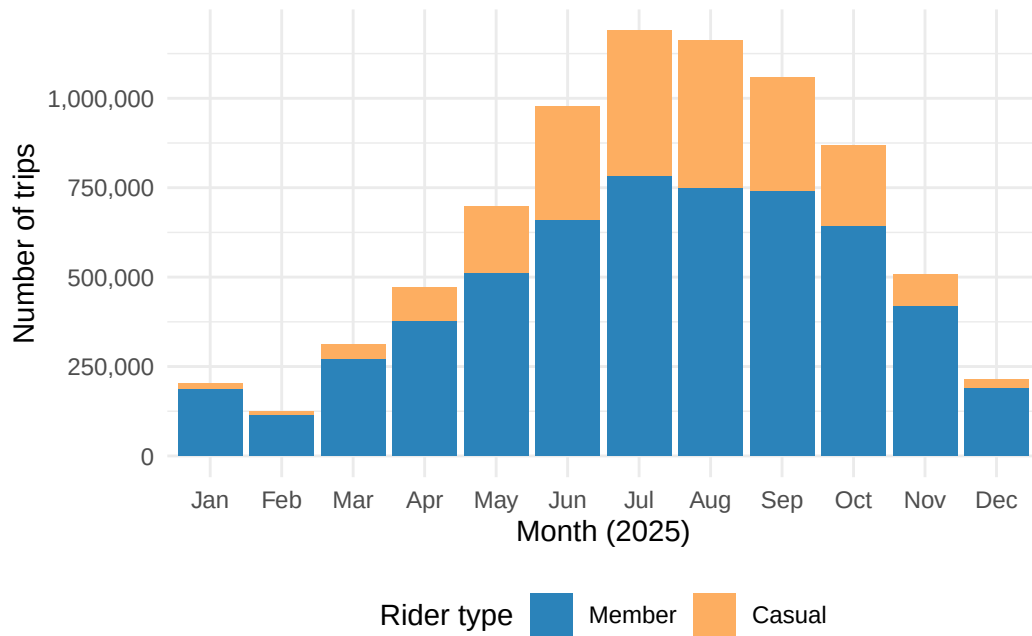


Figure 1: Number of Bike Share Toronto trips in each month of 2025, split by rider type. Casual trips range from about 9,000 in February to over 400,000 in August, which is from 7% to 35% of each month's trips. Member trips range from about 116,000 in February to approximately 784,000 in July.

July, casual trips grew about 45 times, whereas member trips grew only about 7 times. In the quietest winter months of December, January, February, members still rode approximately 15-25% of their peak months, but casual riders only made about 2-6%. Although our data does not record weather, it does fit evidence that cold and wet weather reduces overall bikeshare use (Gebhart and Noland 2014), affecting casual users more than members (Fishman 2016). Because the data counts trips rather than riders, I cannot show whether winter led to fewer casual riders, or the same users riding less.

2.4 When trips start

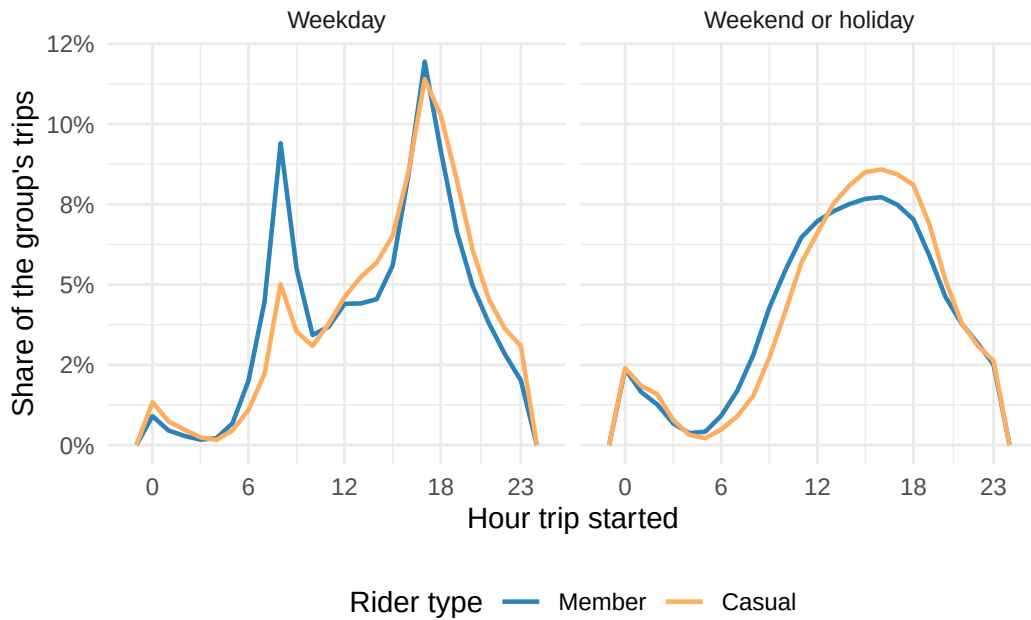


Figure 2: Share of each rider type's trips starting in each hour of the day, on weekdays and on weekends or holidays. On weekdays, member trips peak at 8 am and 5 pm, while casual trips have a smaller 8 am peak and a main peak at 5 pm. Between 7 and 10 am, 19% of member trips and 11% of casual trips start. Between 4 and 7 pm, about 30% of each group's trips start. Both groups peak in mid-afternoon on weekends and holidays.

Figure 2 shows the hour when trips start, as a share of each group's trips. Following El-Assi et al. (2017), I grouped weekends and statutory holidays in Ontario together.

On weekdays, for members, there are two peaks, around 8 am and 5 pm, which is consistent with the average commuting schedule. This is the same pattern seen across 40 cities in Bean et al. (2021). Casual riders display a much smaller morning peak. About 11% of weekday trips

start between 7 and 10 am, much smaller than the 19% for members. However, in the evening, between 4 and 7 pm, both members and casual riders show almost identical activity of around 30%. The evening peak does not distinguish commuting from leisure, the end of the school day or workday leads to both commuting and leisure trips, so similar evening peaks for members and casual riders does not mean that casual riders are commuting.

On weekends and holidays, both groups share a broad afternoon peak. The groups do differ on the amount they ride on these days. Because there are about 2.5 times as many weekdays compared to weekends and holidays, I compare trips per day. Members made 0.77 trips on a weekend for every trip on a weekday, whereas casual riders made 1.45. Just like El-Assi et al. (2017) found in 2013, members tend to ride more on weekdays while casual riders tend to ride more on weekends.

2.5 How long trips last

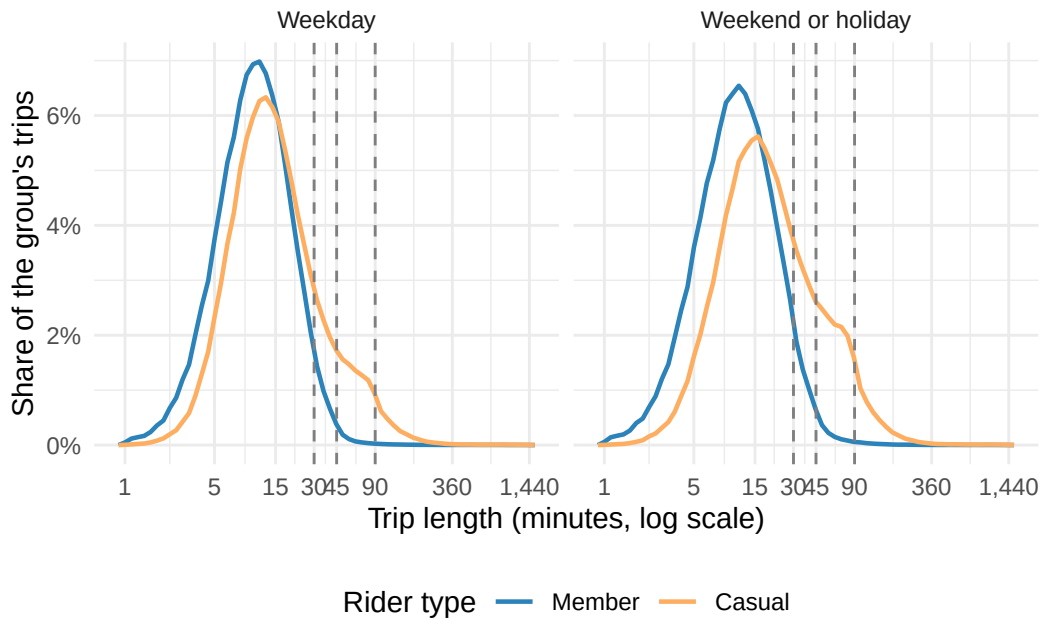


Figure 3: Share of each rider type's trips by trip length in minutes (log scale) on weekdays and on weekends or holidays. Dashed lines mark 30, 45, and 90 minutes. Member trips are concentrated at shorter lengths and fall close to zero beyond 45 minutes. Casual trips peak at slightly longer lengths and have a longer tail, which drops at 90 minutes on weekends and holidays. Throughout 2025, 4.5% of member trips and 22.7% of casual trips last more than 30 minutes.

In Figure 3, the trip length ranges from 1 minute to 24 hours. Most trips only last 5 to 20

minutes, so if we used a normal axis, most trips would be compressed to the very left of the graph. Therefore, I decided to use a log scale to space the axis by multiplication, so that the distance from 1 to 10 minutes now equals the distance from 10 to 100 minutes. Each line displays the share of a group’s trip at every trip length. The dashed lines mark 30, 45, and 90 minutes.

Casual trips are generally longer and more spread out on the graph, even more so on weekends. The median trip length for members is 10.2 minutes on weekdays and 10.7 on weekends and holidays. The median trip length from casual riders is 13.8 minutes on weekdays, and 17.8 on weekends and holidays. Median here means that when trips are sorted by length, half of the trips are shorter than this value, and half are longer.

The pricing tiers for members show up inherently in the figure. There are large drops in member rides at 30 and 45 minutes specifically, which are the two bike session limits in the membership plan. Just 4.5% of member trips exceed 30 minutes, and 0.9% exceed 45. For casual trips, 22.7% exceed 30, and 13.7% exceed 45.

Longer and more variable casual trips fit the idea of leisure riding. This is consistent with Fishman (2016), which reported that casual users generally take longer trips than members. Trip length in our case is measured in time, not distance, so longer trips may include pauses and stops.

2.6 Casual riders across the seasons

Figure 4 shows only weekday trips in winter (January and February) and summer (July and August), grouped by member or casual rider. In the casual panel, the blue winter line shows a clear morning peak at around 8 am, which is less obvious on the red summer line. But for the member panel, both the winter and summer lines nearly overlap the entire time, since members generally ride at the same times throughout the year. In Section 2.4, I discussed how morning hours showed clear signs of commuting. For casual riders, 17% of weekday trips in the winter start between 7 to 10 am, but only 9% of their trips start within that timeframe during the summer. The change for members is a lot smaller, about 21% during the winter and 18% during the summer.

The number of casual weekday trips was 494,552 in the summer, and fell to 18,652 in the winter, which was about 26 times fewer. Casual morning trips fell from 44,859 to 3,192, which is only about 14 times fewer. Since morning trips fell much less than casual riding overall, the remaining casual riding during the winter is more concentrated at commuting times. This suggests that the “casual” tag may cover two groups. One smaller group of “casual” riders ride year-round, and their weekday trips resemble those of members, and a much larger group that mainly ride during the summer, with patterns that look like leisure riding. This pattern is consistent with El-Assi et al. (2017), which found that Toronto’s trips shifted towards rush hours in the winter as casual use fell but members continued to commute. Fishman (2016) observed that bad weather deters casual users more than members.

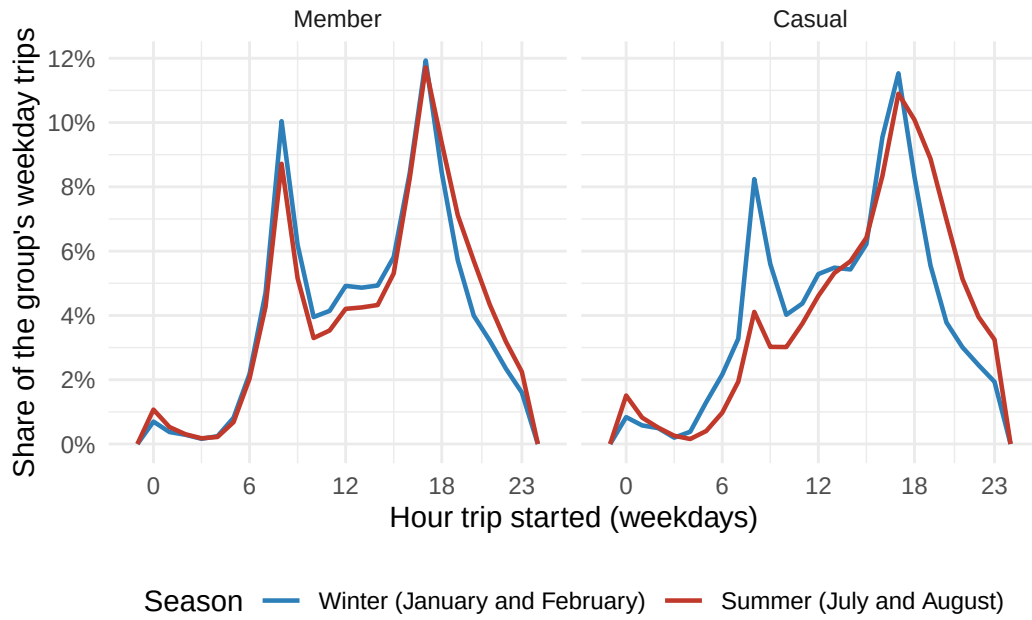


Figure 4: Share of each rider type's weekday trips starting in each hour, in winter (January and February) and summer (July and August). Among casual riders, 17% of weekday trips start between 7 and 10 am in winter, compared with 9% in summer. Among members the share is 21% in winter and 18% in summer.

A possible explanation for this pattern is that residents who pay per ride make up a larger proportion of winter casual riding since fewer visitors are in the city at the time (Section 2.2). However, because the data records trips and not riders, I cannot confirm whether the same casual riders change their riding habits during the winter, or whether different people entirely ride during the winter.

2.7 Summary Statistics

Table 2: Member and casual trips in 2025. Percentages are shares of each group’s own trips, except weekday trips starting 7 to 10 am, which are shares of each group’s weekday trips. Median trip length is across all days.

Measure	Members	Casual riders
Trips	5,654,702	2,134,140
Share of all trips	72.6%	27.4%
Median trip length (minutes)	10.3	15.2
Trips on weekends or holidays	25.6%	39.4%
Weekday trips starting 7 to 10 am	19.4%	10.8%

Table 2 unites the measures I discussed in the data section. I discussed share of trips and seasonality in Section 2.3, weekday morning trips and weekend riding in Section 2.4, and median trip length in Section 2.5.

Appendix

A Data cleaning

Table 3: Cleaning removed 0.3% of the raw 2025 trips. Each step lists the trips remaining after it and how many it removed.

Step	Trips remaining	Trips removed
Raw 2025 trips	7,812,520	
Removed trips missing a start station, end station, or end time	7,805,563	6,957
Removed trips shorter than 60 seconds	7,788,854	16,709
Removed trips longer than 24 hours	7,788,842	12

- Read 12 monthly csv files using Windows-1252 encoding so that station names with dashes could display correctly.
- Removal numbers are given in Table 3. Removals included incomplete trips (bike not docked, lost, stolen), trips under 60 seconds (failed unlocks, El-Assi et al. (2017) used 30 seconds for the same reason), and trips over 24 hours (likely system errors).
- Reconstructed station names for January–October, where end-station names repeated the start-station name. Rebuilt names by using station IDs to match each station’s most recent name. 20 stations only ever appeared as trip ends, and labeled “Station [ID]”.
- Constructed the following variables: trip length in minutes, month and start hour, day type: weekends plus Ontario’s nine statutory holidays in 2025, bike type and round trip.
- Saved data as Parquet files, one per month, so I stay under the 100MB GitHub file limit.

B Data tests

Tests were run for:

- **Structure:** Columns are expected, no missing values, no duplicate trips, and row counts match the cleaning log.
- **Valid categories:** I only see expected rider types and day types.
- **Dates/Times:** All days are in 2025, month and the hour agree with the start time. All holidays are grouped correctly.
- **Duration:** Duration is between 1 minute to 24 hours.
- **Stations:** Names encoded correctly, one name per station variable.

Tests were run and passed on simulated data first before actual data. Check GitHub repository for 01-test_simulated_data.R and 04-test_analysis_data.R for details.

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